Problem 11: Damping

Setup

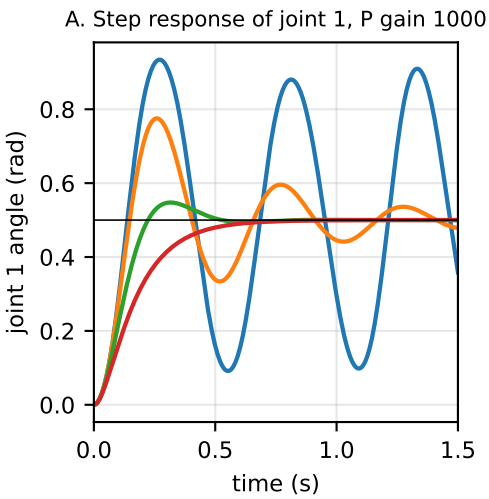
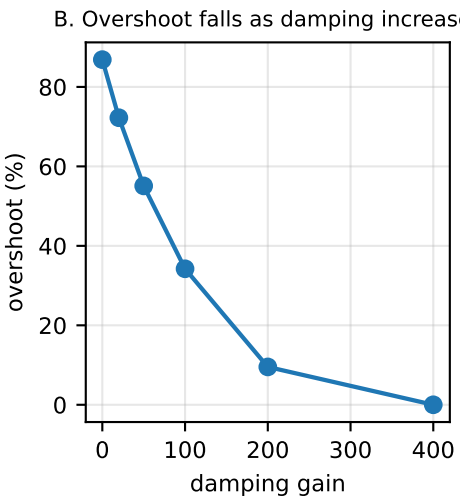
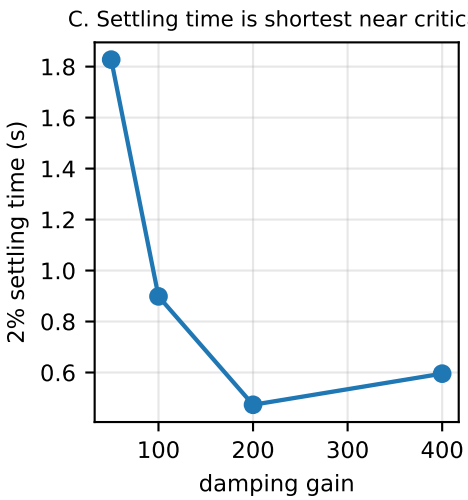
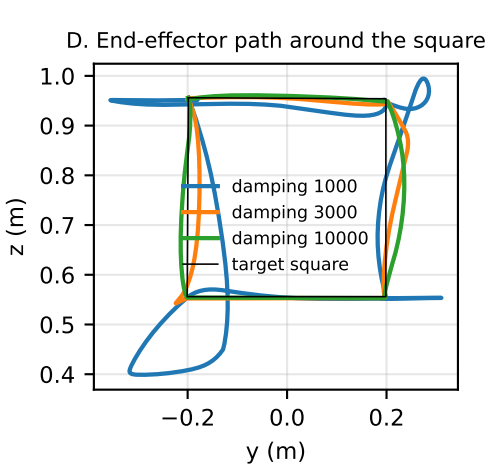
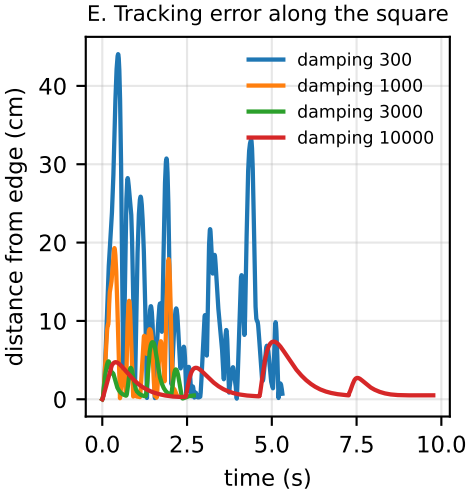
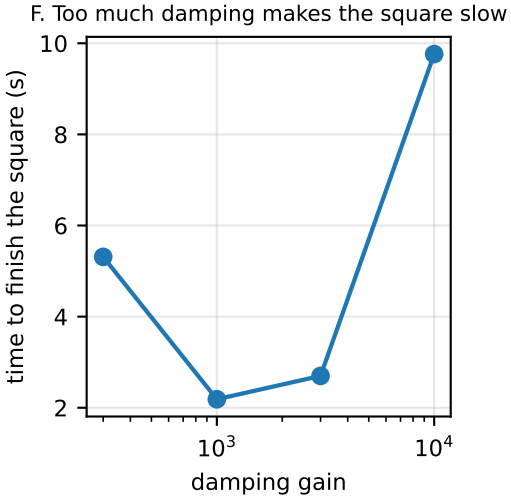
I applied a 0.5 rad step to joint 1 with P gain 1000 and swept the damping gain of the PD controller. I chose joint 1 because it rotates about a vertical axis, so gravity has no effect on it and the response should be a clean second-order one.

Step response

With no damping the joint rings for the entire run (panel A). Adding damping reduces the overshoot and the ringing. At 50 the response is clearly under-damped, at 200 it has only a small overshoot and reaches the target fastest, and at 400 it is over-damped (no overshoot but a noticeably slower approach). Panel B shows the overshoot falling steadily with the damping gain, from 87% with no damping to zero at 400. The settling time (panel C, time to get within 2% of the step and stay there) behaves differently. It drops quickly at first, is shortest at 200 (0.47 s) and then rises again at 400 (0.60 s) because the over-damped response is slow to arrive. Since 200 still overshoots slightly and 400 does not, critical damping for this joint must be somewhere between the two.

Square task

I then drew the square from 9.5 (P gain 10000) at several damping gains. With damping 1000 the arm reaches each corner quickly but overshoots, so the sides of the square bulge outwards (panel D). With 3000 the sides are nearly straight. With 10000 they are straight as well, but the arm approaches each corner so slowly that the whole square takes 9.8 s instead of 2.7 s (panel F). This makes sense because the waypoint follower only moves on once the arm is within 0.01 rad of the current corner, and an over-damped arm creeps towards its target. Panel E shows the tracking error, i.e. the distance of the end effector from the edge it is meant to be following, over time. With 300 and 1000 the error spikes after every corner as the arm overshoots, whereas with 3000 and 10000 it stays within a few centimetres. At 300 the arm is so under-damped that it takes 5.3 s (I left this one out of panel D because it wanders all over the plot, but it is in panels E and F).

Conclusion

Therefore, more damping helps by removing overshoot and giving straight edges, but hurts by slowing down the approach to every waypoint. Past critical damping there is no accuracy left to gain, only time to lose.